Pergamon

*Acta Astronautica* Vol. 46, Nos. 2–6, pp. 365–374, 2000
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Printed in Great Britain
PII: S0094-5765(99)00232-5 0094-5765/00 $ - see front matter

# THE USE OF SMALL SATELLITES IN THE NASA EARTH SCIENCE ENTERPRISE (ESE) EARTH OBSERVING SYSTEM (EOS)

**Larry J. Paxton, Jeng-Hwa Yee, Glen Fountain and C.I. Meng**
The Johns Hopkins University
Applied Physics Laboratory
11100 Johns Hopkins Rd.
Laurel, MD 20723 USA

## ABSTRACT

As NASA looks to reshape the Earth Observing System for the post-2002 mission timelines, we argue that small, commercial buses can be modified to carry key instruments as "hot" spares, technology demonstrations, or in support of new programmatic imperatives. We consider the particular case of the Iridium bus flying a new sensor to demonstrate the utility of measuring ozone via stellar occultations. We find that a very low cost demonstration mission could be mounted that could be launched to fly in association with a planned EOS element, EOS CHEM. This project would serve to validate the new measurement approach and it would demonstrate the idea of launching to replace or update an element in an existing constellation. Such a mission would be a natural follow-on to the PICASSO-CENA ESSP mission. PICASSO-CENA is planned to fly in formation with EOS PM. Our mission would demonstrate that many of the current instruments and measurement requirements addressed by EOS AM and PM could be maintained through use of such small satellites, an issue not fully addressed by PICASSO-CENA or any other ESE mission. © 2000 Elsevier Science Ltd. All rights reserved

## 1. INTRODUCTION

The United States's National Aeronautics and Space Administration (NASA) Earth Science Enterprise (ESE) plans to launch the first of its Earth Observing System (EOS) satellites, AM-1, in Summer 1999. The next satellite follows about two years later (PM-1), with the final member of the EOS constellation being EOS Chem-1 launched in late 2002. The original instrument and satellite constellation was envisioned as consisting of two more satellites (AM-2 and PM-2) yielding a series of extended measurements reaching to the end of the next decade.

The April 15, 1997 baseline showed a plan to continue key environmental measurements for at least ten more years using the same instruments, buses, and system architecture. NASA planners recognized the limitations inherent in this approach and so considered the alternative step: making the large platforms serviceable. While this had obvious technical issues it did indicate that there was a plan in place to maintain continuity of those key parameters required to baseline the state of the atmosphere and to address the roles of anthropogenic influences in changing that baseline state that could, in principle, work.

| Mission Launch Date | Follow-on Launch Date |
|---|---|
| TRMM 6/97 | TBD/ATMOS-A |
| CERES inclined-orbit FOO 2000 | |
| ACRIM 1998 | TSISat 2001 |
| AM-1 6/98 | AM-2 6/04 |
| Landsat-7 9/98 | |
| SAGE-3 on Meteor 6/98 | SAGE-3 on Station 2001 |
| | SAGE-3 inclined-orbit FOO 2005 |
| ADEOS | ADEOS-2/Seawinds 2/99 |
| Jason-1/Radar Alt 1999 | Radar Alt-2 2004 |
| PM-1 12/00 | PM-2 12/06 |
| | NOAA-N 2007 |
| Laser Alt-1 7/02 | Laser Alt-2 7/07 |
| SOLSTICE 12/02 | SOLSTICE FOO 2008 |
| CHEM-1 12/02 | CHEM-2 Monitor (includes SAGE-3) 6/08 |
| | CHEM-2 Process 12/08 |

Table 1. The "Old" EOS Plan.

In this paper we have briefly considered the issues raised by the increased availability of small commercial satellite buses, reliable small expendable launch vehicles, and the rapid growth in data processing capabilities. While data users are always leery of the next "big thing", planners must be wary of pursuing technology as a panacea. Appropriate technology, preferably developed at some one else's expense, should be incorporated as a mission enabler. The size and complexity of a satellite bus is driven by the telemetry and pointing requirements as well as the size of instruments. Spacecraft subsystems are subject to technological advances that have greatly reduced their size and power requirements. Optical sensors are, however, driven by their "$A\Omega$" (the product of the effective collecting area and the field-of-view): a certain number of photons must be collected in order to make the measurement. To do that quickly, for a given ground sample distance, the aperture must be of a certain irreducible size. Optical systems may not decrease in size but their structures could become an integral part of the spacecraft, further reducing the need for a massive, stiff bus. New approaches will surely evolve over the next few years as the technology required to mount deep space missions drives satellite subsystem designs in new directions. The commercial communications industry is driving the next generation of rad hard processors which will enable on-board data reduction and targeted data product generation: further reducing the size of spacecraft by reducing communications requirements and the subsystems needed to support that function.

We propose that a small instrument payload could, and should be launched to demonstrate the use of a small commercial bus. Our candidate is the STARS sensor which could fly in concert with the EOS CHEM mission. Compelling science reasons exist but programmatic ones are just as easily

found: demonstration of a new instrument concept will lay the ground work for flight of replacement sensors using the same bus. With this ability comes a provision for data continuity that has yet to be addressed by NASA's post-2002 plans.

## 2. THE EARTH OBSERVING SYSTEM ARCHITECTURE AND ISSUES

The current EOS architecture shows plans for missions out to the year 2003. The original EOS mission suite was designed to provide a continuous set of comprehensive data products for 15 years. A key element in the EOS concept was the EOS Data Information System (EOSDIS) which was intended to produce data products for all these instruments and distribute those data products to the larger community. The scope of the problem was, and remains, enormous. There were, of course, a number of problems with the original EOS mission concept that have been, for the most part, remedied in whole or part by new activities established within ESE. Some of those problems are immediately obvious: the investigations were chosen and the instruments selected twenty years before the launch of the last component, there was no provision for new technology, the platforms were enormous and complex and their associated technology had to be supported over the entire span of the mission, no provision could be made for replacing a sensor if it failed, the plan envisioned a fifteen year data set with these same instruments, there was no plan for validating new technology on orbit, and because of the size and complexity of the platforms many areas relevant to ESE could not be easily incorporated into the mission plan. The initiation of the Earth System Science Pathfinder (ESSP) program was a bold and necessary step toward recovering some of the "missing" science. With the latest addition, the Earth Observer (EO) series, new technology can be tested and evaluated for incorporation into new missions. Data from other non-NASA satellite programs, such as the Defense Meteorological Satellite Program (DMSP) and the National Oceanic and Atmospheric Administration (NOAA) system or polar and geosynchronous satellites, are now viewed as partners in the collection of environmental data. Commercial sources and partners are now a part of the mix with data buys, such as the instituted for the SeaWiFS program, a component of many programs. With this past record of accomplishment in reshaping the EOS program and the ESE activities, there remain areas for further work in delineating new architectures that respond to evolving needs while making maintaining a continuous record of observations.

## 3. NEW APPROACHES AND ISSUES

One of the earliest "requirements" cited for the large size of the EOS platforms was that of "simultaneity". That is to say, that the Earth system had such a large amount of inherent variability that a measurement days or hours later was of little use. At the time, some ten years ago, a single large platform seemed the only viable means of making many different kinds of measurements at the same time. Formation flying is now often considered as a way of achieving simultaneity with many instruments on different spacecraft. Command, control and communications become more complicated, of course, but there are

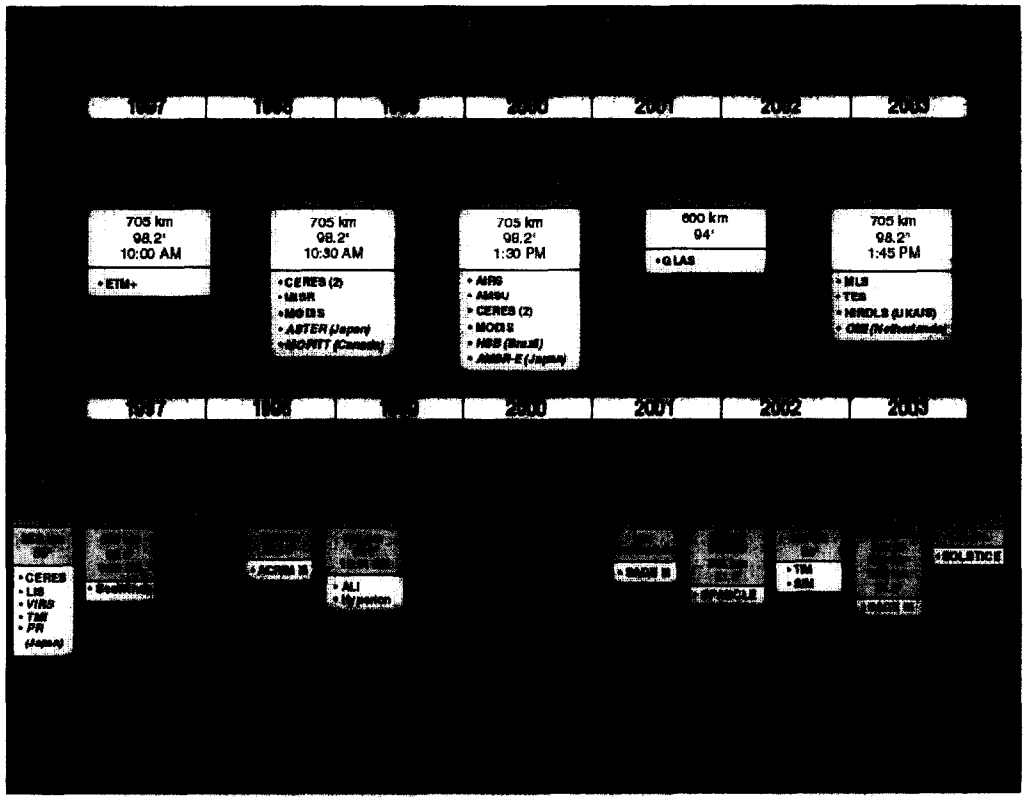



Figure 1: Pre-RFI Mission Scenario. Taken from NASA ESE Web site (http://eospso.gsfc.nasa.gov/Images/mission.gif).

compensating factors. Principle among them is the growing, competitive market for producing small spacecraft. This, in itself, is a development that would have been difficult to foresee fifteen years ago. The paradigm then in existence was that basic research was the technology driver for spaceflight (just as it was held that basic research was the driver for computers and we know where that has led: most of us use "personal computers" even for advanced data processing). We propose here the "Winnebago" model for spacecraft buses. Winnebago is a commercial manufacturer of motorhomes or caravans. These vehicles are constructed on a commercially available truck or van chassis which provides the engine and drive train components. The vehicle is available with a number of "options" that can be tailored to the individual's needs and so are able to accommodate almost all users: the market share for fully custom one-off designs is very small. This seems an apt analogy: a small commercially available bus could, in principle, be modified to accommodate a broad range of possible uses yet still exhibit substantial cost savings over a custom-built low-volume spacecraft design.

Commercial telecommunications satellites are produced in relatively large production runs that lead to relatively low costs. It is not unusual to see a commercially available system with an on-orbit power of 1 kW and an effective payload of hundreds of kg. These satellites have the further advantage of being adapted to more than one launch vehicle and launch site. From one to several satellites can be placed in orbit at one time. This allows a great deal of flexibility in terms of populating the constellation and for replenishing or augmenting an existing configuration.

Remote sensing experiments typically require fairly precise pointing (typically about 0.03°, at the finest). The costs of augmenting the "off-the-shelf" attitude control system are manageable, however. We also note that non-recurring engineering costs can be spread out over all of the modified buses that have similar pointing requirements thus further reducing the effective cost for the augmentation of these buses. In the past, the use of a commercial satellite bus has often not been particularly attractive financially because many modifications are required to the one and only such platform, thus rendering this option virtually indistinguishable from a custom one-off bus.

There are, of course, a number of problems associated with maintaining a number of spacecraft in the same orbit. Fortunately, none of the instruments on the EOS AM or PM platforms or currently slated for flights of opportunity require precise station keeping: knowledge rather than control is the driver. Commanding and data downlink present some challenges. Commanding is fairly simply handled by techniques currently used to prevent the inadvertent operation of one spacecraft by commands intended for another. Data downlink is more of an issue since all elements of the constellation are in sight of a single ground station for about the same period each overflight. There are simple and complex remedies to this problem. The simplest and least elegant remedy is to increase the downlink rate and/or the number of stations capable of receiving the data so that one satellite's data can be downlinked at a time. Onboard storage requirements are actually reduced by separating instruments: flight of a mission's payload on a number of spacecraft actually enables one to perform data downlink load balancing wherein the optimal partitioning of data streams can be structured.

### 3.1 Iridium as a Paradigm for a Future Small Satellite Mission Bus

The Lockheed Martin Missiles and Space Iridium bus is capable of supporting almost all of the EOS baselined sensors. The LM700 bus comes in two varieties, and "A" and a "B" format. The "A" format resembles most closely the bus used for the Iridium constellation whereas the "B" variety has the attitude & control payload sensor module relocated within the composite structure. The estimated costs for a bus taken from the production line are very low (depending on the options, between $4M and $10M). This, combined with a flexible modular design would seem to make them ideal for flying replacement sensors for an aging EOS bus or to fly new sensors as a proof of concept.

| capabilities | LM-700A (B) | LM-900 |
|---|---|---|
| propulsion: | monopropellant (N2H4) | monopropellant (N2H4) |
| | nom. del-V = 1690 ft/s | total impulse = 16,700 lbf-s |
| attitude control: | 3-axis stable | 3-axis stable (zero momentum) |
| | jitter < 100 microrad | ultra low jitter (?) |
| | control = 0.4 deg (R, PI, Y) | control = 12 arc sec (1-sigma) |
| | know = 0.15 deg (R), 0.3 deg (PI, Y) | highly agile; 4 deg/s, 0.2 deg/s^2 |
| electric power: | 500 W orbital avg., EOL | 1250 W BOL |
| | 28 +/- 6 VDC unregulated | 28 +/- 6 VDC unregulated |
| | <0.5 V (pk-pk) ripple | sgl-press vessel NiH2, 50A-hrs |
| thermal control: | electronics pallet, 500 W heat rejection capability | precision thermal control available (?) |
| on-orbit design lifetime: | 7 years | > 7 years, MMD > 5 yrs |
| payload weight: | 500 lbs (can be <700 lbs) | < 1000 lbs |
| payload volume: | electronics (internal) = 63" x 19" x 11", sensors/antennas = 65" x 26" x 18" nadir sensors = 40" dia. x 35" long | electronics (internal) = 31" dia. x 40" external volume = 87" dia. x 50" electronics bay = 37" x 33" x 15" |
| FOV: | hemispherical centered on nadir vector, hemispherical centered on velocity vector | hemispherical centered on nadir vector |
| EMC: | qualified to MIL-STD-461/462 | (?) |
| | low-noise, low-radiated emission power bus, shielded enclosure for payload electronics (OPTIONAL) | |
| on-board storage: | 1 to 64 Gbyte solid-state storage (OPTIONAL); 320 MHz I/O | 75 Gbits (BOL) EDAC-protected high-speed memor |
| communications: | STDN S-band, SGLS system, link security & authentication feature (OPTIONAL) | STDN S-band, SGLS system, link security & authentication feature ; Wideband capability (OPTIONAL ?) X-band, 320 Mbits/s downlink, high gain 2-axis gimbaled antenna |
| C&DH: | Space-qual 32-bit processor, 5.25 -to- 9.00 MIPS throughput, 5V and 26V command channels, MIL-STD-1553B interface, 0 -to- 5 V bi-level and conditioned telemetry, 400 Kbytes PROM & 1 -to- 3.0 Mbytes RAM. | Space-qual 32-bit RISC processor, 12 MIPS throughput, MIL-STD-1553B interface, serial & bi-level command channels, bi-level & conditioned telemetry channels. |
| GPS: | (?) | GPS P-code L-1 omni antenna & RX. |
| LV accommodations: | fits fairings for: LMLV-1 (only B-variant), LMLV-2, LMLV-3, Proton, Atlas II, Delta, Long March, weight = ?, triangular configuration. | fits 87" ID (92" OD) LMLV fairing, weight = 1100 lbs, hexagonal configuration |

Table 2. Parameters definig the LM700A and LM700B buses.

## 4. ADVANTAGES OF A CONSTELLATION

Implementing the EOS payloads on small satellites does have significant advantages. Chief among them is flexibility in configuration. That flexibility is realized in two powerful areas: first instrument designs can evolve while maintaining a baseline and second new measurement concepts can be implemented. For example, ozone measurements are required in order to track changes in the photochemical properties of the atmosphere as well as to understand and document the efficacy of mitigation processes such as the Montreal Protocols. These measurements have suffered from periodic interruptions in the measurement history due to instrument failures and programmatic issues which have prevented replacement sensors from being baselined against the earlier sensor. When small decadal trends are to be tracked this presents a real problem. New measurement techniques can either extend the temporal or altitude coverage of existing measurements but often lack a suitable

platform at a sufficiently low cost. By implementing a strategy wherein new instruments can join existing ones in orbit, concerns about simultaneity or common volume sampling can be reduced or eliminated. This improves the final product and increases the fidelity of the measurement and allows absolute measurement techniques to provide a calibration for relative techniques. New instrument concepts are being developed which will provide tropospheric ozone. Common volume measurements of total column ozone and tropospheric ozone would serve to delineate the photochemistry and transport of pollutants on regional and super-regional scales. Programs, such as ESSP, could be incorporated directly into the EOS constellation (where appropriate) by adding that new ESSP satellite to the constellation. The ESSP payload could be flown along with a replacement sensor or a sensor upgrade for the EOS instrument suite. For example, Pathfinder Instruments for Cloud and Aerosol Spacebourne Observations - Climatologie Etendue des Nuages et des Aerosols (PICASSO-CENA), which makes measurements of the aerosol and cloud vertical distributions, aerosol extinction, optical depth, and single scattering albedo, cloud extinction, optical depth, and emissivity, cirrus asymmetry parameter and effective particle size, surface and atmospheric radiative fluxes would be an excellent compliment to the EOS payloads (either AM/PM or CHEM) and currently is slated to fly in formation with EOS PM-1.

Measurements can also be extended by flying on more than one platform. For instance, the Multi-angle Imaging Spectroradiometer (MISR) science return would be enhanced by the ability to make simultaneous common volume measurements. With just one MISR the common volume is sampled at different times for different angles. Two MISR on spacecraft flying in formation could eliminate that problem and reduce the complexity of the instrument by reducing the number of angles that had to be accommodated by the sensor head. The Advanced Spaceborne Thermal Emission and Reflection Radiometer (ASTER) makes high spatial resolution thermal IR measurements over a fairly narrow swath which leads to a long revisit time for mid- and low-latitude observations. Multiple spacecraft flying in formation can extend either the local time coverage of the measurement or the cross-track coverage at a fixed local time.

### 4.1 Implementation of a Constellation

There are, of course, many issues concerning the implementation of a constellation of satellites operating in close proximity. Whereas one can imagine how this might be managed, the mission operations costs would seem certain to overwhelm any cost saving realized by the use of small satellites. Autonomous operation, a requirement for the latest generation of deep space missions, holds the promise of significant reduction in operations costs. This would be achieved by implementing orbit control on-board the spacecraft.

The software to do this has been under development by commercial concerns as well as government agencies. Primary functions include constraint resolution, on-board orbit propagation, and reduction in the need to uplink commands and/or orbit solutions. Precise formations, dynamically adjusted for sun illumination or geographical region of

coverage, could be envisioned. This would enable new investigations to be performed or to reduce instrument complexity e.g. by autonomously "nodding" the spacecraft to avoid a glint-viewing geometry a requirement on instrument design could be reduced or eliminated.

### 4.2 Technology Drivers and Solutions

The quest for solutions to the problems inherent in flying space missions exists in a societal and technical matrix that is profoundly different than that which existed when the Earth Observing System was first conceived. Access to space has been diversified among many countries and is now a "commercial" enterprise rather than an exercise of national "will". Science and defense are rarely the drivers for new technology: neither effort can afford the magnitude of expenditure and risk generally accepted as necessary in today's business environment.

Consider the case of the Pentium chip and how far we have come in terms of the evolution of microprocessors. The IBM PC was introduced in 1981 and featured a 4.77 MHz processor. The original PC was equipped with 16 kilobytes of memory, expandable to 256 k, and one or two 160 k floppy disk drives. Prices started at $1,565 US. Currently one can buy a Pentium III system which is about a factor of a thousand faster, uses 32 bit words, and at least 10,000 times more memory for the same price. These incredible changes (commonly encapsulated under the rubric "Moore's Law: Processor speed, memory density, disk drive size, etc. doubles every 18 months) provide the imperative for designing a flexible spacecraft architecture. Space systems have not quite kept pace with this growth rate because of the lack of an economic imperative: until recently the market for high-speed rad hard processors has been too small to be worth pursuing and no commercial advantage was seen in doing so.

Intel, the US Air Force Research Laboratory (AFRL), and Sandia Laboratories have just announced that they have formed a consortium to develop a "rad hard" version of the Pentium microprocessor. Under the newly announced arrangement Intel will supply Sandia with the design for the chip. Intel estimates that the chip design represents an investment of $1B in commercial funds placing such an effort outside of the realm of any basic research or defense activity. In return, Sandia and AFRL will supply funds to produce a rad hard version of the chip. This will yield at least a factor of ten improvement in processing power and will undoubtedly lead to increases in memory density and volumes akin to those we have seen in terrestrial applications.

## 5. CONCEPT DEMONSTRATION

The STARS instrument (discussed by Yee et al. in these proceedings) would be an ideal candidate for such a flight demonstration.

STARS could be launched to join with the EOS CHEM satellite as early as 2003. This would fulfill many programmatic requirements including: demonstration of augmentation or replacement of elements of a sensor suite, infusion of new technology into a program, cross-validation of a new technique and performance against other baselined sensors, as well as demonstrating "hand over" from one measurement set to the next. This would

be a major step in demonstrating the ESE's EOS program's ability to meet data continuity requirements.

### 5.1 Implementation

STARS is a self-pointing instrument which autonomously acquires and tracks stars as they set through the Earth's lower atmosphere. STARS uses hyperspectral imaging of a stellar spectrum to determine the amount of absorption along the line-of-sight. The method is self-calibrating since it measures the spectrum of the star as seen above the Earth's atmosphere as a reference before the occultation begins. Changes in the observed stellar spectrum, as a function of wavelength, are readily converted into key, difficult to measure, constituent densities. Ozone and water vapor are among the most important of these constituents.

STARS builds upon solar occultation techniques but has the advantage of viewing a point source compared to an object (the Sun or moon) with an apparent "size" of about 30 km. STARS also measures ozone at night rather than at sunset or sunrise when the atmospheric chemistry is being driven by changing solar inputs. STARS obtains about 10 times more profiles than a solar occultation instrument.

STARS is complementary to the EOS CHEM mission as developing an ozone climatology is one of that missions major goals. NASA ESE is also seeking ways to monitor very small decadal time-scale changes in ozone and there a self-calibrating technique has many advantages.

## 6. SUMMARY AND CONCLUSIONS

In summary, we find that the basic measurement requirements of the EOS platforms can be met by flying a few small satellites, based on commercially available buses, in formation. While there are some complications which arise from such an architecture, the advantages are enormous as they insure continuous high-quality measurements that respond to technological advances and scientific push.

In this paper we have briefly considered the issues raised by the increased availability of small commercial satellite buses, reliable small expendable launch vehicles, and the rapid growth in data processing capabilities. next generation of rad hard processors which will enable on-board data reduction and targeted data product generation: further reducing the size of spacecraft by reducing communications requirements and the subsystems needed to support that function.

We propose that a small instrument payload could, and should be launched to demonstrate the use of a small commercial bus. Our candidate is the STARS sensor that could fly in concert with the EOS CHEM mission. Compelling science reasons exist but programmatic ones are just as easily found: demonstration of a new instrument concept will lay the ground work for flight of replacement sensors using the same bus. With this ability comes a provision for data continuity that has yet to be addressed by NASA's post-2002 plans. Under an architecture such as we have briefly considered here, small satellites, based on commercial buses, would be launched into orbit to join an existing EOS platform. Once there the old sensor and its replacement would be cross-correlated and inter-calibrated. As new instrument concepts were introduced they, too, could join the constellation. If the key environmental parameters derived from

the new and old sensors were determined to be the same or that an improvement was noted and understood, then "hand-over" would have been achieved. Small satellites would also provide opportunities for other single, prototype sensors (or associations of sensors) for measurements such as ocean color, ocean surface winds, ocean height, radio occultation sounding, and ozone that could benefit from additional measurements that, while useful, may fill a correlative or corroborative role. Small satellites carrying specific instruments could also be used to respond to new and evolving measurement requirements or other "service" activities such as hazard monitoring and warning without impacting the operation of a research platform that they are flying in association with.